

ARUN SANYASI

Instrumentation and Control Engineer

PLC/SCADA | Electrical Automation | Engineering Operations | Facility Automation | Commissioning

Samdrup Jongkhar, Bhutan | +975 17914131 | sanyasiarun220@gmail.com

Website: <https://arunsanyasi2786.github.io/arun-portfolio/>

CAREER OBJECTIVE

Graduate Instrumentation and Control Engineer seeking a growth-oriented engineering operations, automation, maintenance, facility support or commissioning role where I can apply PLC/SCADA knowledge, electrical systems exposure, maintenance coordination, technical documentation, team communication, safety awareness and service-focused problem solving to support reliable plant and facility operations.

PROFESSIONAL PROFILE

Instrumentation and Control Engineering graduate from the College of Science and Technology, Royal University of Bhutan, with practical exposure in hydropower automation, ferro-alloy plant electrical systems, PLC/SCADA integration, HT/LT panel assembly, commissioning support, field instrumentation, facility automation concepts and operations documentation.

Experienced with Siemens SICAM A8000 controller modules, SCALA 250 SCADA screens, EPLAN electrical schematics, signal tracing, ferruling, labelling, field wiring verification, gate-control logic, switchyard systems, MCCs, motor-control wiring, transformer protection systems, soft starters, VFDs and preventive maintenance observation.

The profile also includes leadership exposure as Class Representative, Public Address Councilor, student project contributor, business idea competition winner and founder of a virtual reality venture. Additional training in management leadership, principles of management, project management, business productivity and entrepreneurship supports readiness for roles requiring coordination, reporting, follow-up and operational discipline.

KEY SKILLS SUMMARY

Engineering Operations	Task coordination, technical reporting, maintenance follow-up, documentation control, safety awareness and team communication.
Facility and Service Support	HVAC control concept, CCTV connectivity support, equipment checking, preventive maintenance records, safety monitoring and operational reliability.
PLC/SCADA and Automation	Siemens SICAM A8000, Siemens S7-1200, SCALA 250, TIA Portal, WinCC, Factory I/O, CODESYS, FluidSIM, ladder logic and HMI logic.
Electrical and Maintenance	HT/LT panels, MCCs, ACB/MCCB, busbars, motor-control wiring, capacitor banks, soft starters, VFDs and 33kV/11kV exposure.
Instrumentation	RTDs, thermocouples, flow monitors, pressure transmitters, field sensors, instrumentation loops and sensor-to-panel signal tracing.
Leadership and Management Readiness	Class representation, public communication, team coordination, project planning awareness, stakeholder communication and service-focused discipline.

CORE ENGINEERING STRENGTHS

Hydropower Automation	Inlet, outlet and flushing-gate control logic, SCADA monitoring, signal mapping, remote/local operation, field verification and commissioning support.
Control Engineering	Ladder logic, structured text, PID control, interlocks, alarms, safety conditions, sequencing, auto/manual operation and HMI logic.
Commissioning Support	Loop checks, I/O verification, point-to-point checks, signal tracing, ferruling, labelling, field wiring verification and site documentation.
Protection Awareness	Differential protection, overcurrent, undervoltage, overvoltage, earth fault and restricted earth fault protection exposure.
Industrial Plant Exposure	Furnace systems, 18 MVA furnaces, RMHS, PLC, GCP, MCC, hydraulics, pumphouse and transformer systems.
Documentation	EPLAN drawing interpretation, commissioning notes, loop-check records, maintenance observations, task reporting and handover documentation.

EDUCATION

2022–2026	B.E. in Instrumentation and Control Engineering College of Science and Technology, Royal University of Bhutan	68.21% Till 7th Semester
2020–2022	Higher Secondary Education, Class XII Karmaling Higher Secondary School	70.2%
2019	Secondary Education, Class X BCSE	78.14%

PROFESSIONAL AND INDUSTRIAL EXPERIENCE

Bhutan Automation and Engineering Limited

12 Weeks

Graduate Engineer Trainee – Automation, SCADA and Commissioning Dagachhu Hydropower Project / Tala Hydropower Panel Works

- Completed 12-week On-the-Job Training under projects associated with Druk Green Power Corporation.
- Gained field exposure at Dagachhu Hydropower Project and panel assembly exposure for Tala Hydropower Project.
- Worked in live hydropower automation activities involving control panels, protection systems, SCADA integration and commissioning.
- Interpreted EPLAN electrical schematics and supported signal tracing, signal identification, ferruling, labelling and field wiring verification.
- Assisted installation and configuration of Siemens SICAM A8000 controller modules including AI, DI, DO and AO modules.
- Developed SCADA screens with signal mapping for equipment monitoring, visualization and operational support.
- Developed inlet, outlet and flushing-gate logic covering manual, automatic, remote and local operating modes.
- Applied interlocks, safety conditions and operational sequencing in hydropower gate-control logic.
- Supported testing, validation and field verification of developed logic in coordination with the engineering team.
- Contributed to HT/LT panel assembly including ACB/MCCB installation, internal wiring, breaker panel assembly and busbar installation.
- Assisted mini test bench verification, uploading of controller cards and final inspection support.
- Followed wiring discipline, mechanical alignment, safety clearances, operational procedures and site safety practices.

Druk Wang Alloys Limited

06 Dec 2024 – 03 Jan 2025

Instrumentation and Electrical Automation Intern

Electrical Department, Ferro-Alloy Plant

- Completed one-month internship under the Electrical Department at Druk Wang Alloys Limited.
- Introduced to switchyard, furnace systems, HT/LT substations, RMHS, PLC, GCP, MCC, hydraulics and pumphouse.
- Gained exposure to transformer protection schemes including differential protection, overcurrent, undervoltage, over-voltage, earth fault and restricted earth fault.
- Introduced to PLC and SCADA systems of two 18 MVA furnaces and basic industrial automation applications.
- Familiarized with RTDs, thermocouples, flow monitors, pressure transmitters and field sensors used throughout the plant.
- Assisted the electrical team in maintenance tasks and daily plant activities.
- Studied latching DOL, Star-Delta and Forward-Reverse motor control wiring concepts.
- Observed soft starters and VFDs used in industrial motor-control applications.
- Built practical understanding of electrical maintenance, plant automation, field instrumentation and heavy-industry process systems.

Fuzzy Automation

Jul 2025

Automation and Digitization App Intern

Digital Automation / Facility Support

- Supported handwritten bills and receipts digitization workflow, testing, digital archiving and structured records.
- Assisted with detection review, source-image handling, structured storage and searchable digital record keeping.
- Contributed to capture-to-storage workflow validation, device configuration, basic networking and troubleshooting.
- Assisted hotel CCTV connectivity, device checking and basic network troubleshooting for surveillance readiness.
- Supported equipment checks, task reporting and service-focused follow-up for reliable daily operations.

SELECTED ENGINEERING PROJECTS

Basic Health Monitoring Station – Final Year Major Project

Developed a multi-parameter health-monitoring station measuring height, weight, BMI, pulse rate, SpO₂, respiratory rate and body temperature. Integrated biomedical sensors, embedded modules, microcontroller/Raspberry Pi workflow, GUI interface and QR-based health-report generation.

HVAC System Automation and Control

Prepared HVAC control concept covering temperature monitoring, airflow monitoring, fan operation, VFD control, sensors, alarms and comfort-focused automation. Applied instrumentation and control principles to facility comfort, reliability, energy awareness and operational safety.

PLC Based Dam Water Level Indicator and Control System

Developed PLC-based dam water-level monitoring and control logic with level indication, alarm response and control decision functions. Applied PLC/SCADA principles to hydropower and water-resource control applications.

Siemens SICAM A8000 Industrial Steam Boiler Monitoring

Developed SICAM A8000 and SCALA 250 monitoring logic for industrial boiler temperature, pressure and level loops, including process visualization, alarm display and operator-interface requirements.

Automated Manufacturing Line using Factory I/O and CODESYS

Simulated a bottle-production workflow covering conveyor movement, sorting, filling, capping, storage and HMI-based operation. Practiced PLC sequencing, sensor-actuator logic, process simulation and troubleshooting.

Other Technical Projects and Case Studies

Designed smart vehicle accident detection concept using sensors and microcontroller logic for emergency alert support. Completed case studies on solar microgrid with load management, AI-driven EV battery management and solid-state transformers for grid modernization.

TECHNICAL SKILLS

PLC/SCADA	Siemens SICAM A8000, Siemens S7-1200, SCALA 250, TIA Portal, WinCC, Factory I/O, CODESYS, FluidSIM
Control Programming	Ladder Logic, Structured Text, PID control, interlocks, alarms, auto/manual logic, remote/local modes, sequencing and HMI logic
Commissioning	Loop checks, I/O verification, signal tracing, signal identification, ferruling, labelling, field wiring validation and point-to-point checks
Electrical Systems	HT/LT panels, 33kV/11kV exposure, MCCs, ACB/MCCB, busbars, motor-control wiring, capacitor banks, soft starters and VFDs
Motor Control	Latching DOL, Star-Delta, Forward-Reverse control wiring, soft starter operation and VFD operation
Instrumentation	RTDs, thermocouples, flow monitors, pressure transmitters, field sensors, control panels and instrumentation loops
Facility Systems	HVAC control concept, CCTV connectivity support, sensors, alarms, equipment checks, maintenance documentation and operational reporting
Protection Systems	Differential protection, overcurrent, undervoltage, overvoltage, earth fault, restricted earth fault, SIPROTEC, ABB Relion and GE Multilin
Protocols	Modbus RTU/TCP, PROFINET, PROFIBUS and basic industrial networking
Design and Software	EPLAN Electric P8, AutoCAD Electrical, Proteus, SolidWorks, Fusion 360, Arduino IDE, Python, C, C++ and MS Office Suite

WORKSHOPS, CERTIFICATIONS AND TRAINING

- BUS401: Management Leadership – Saylor University, 37 hours, Grade: 92.31%.
- Principles of Management – MindLuster Certificate of Achievement.
- Project Management Short Course – Oxford Home Study Centre / Brentwood Open Learning College.
- Hydropower Automation, SCADA and Commissioning OJT at Bhutan Automation and Engineering Limited.
- Industrial Electrical and Automation Internship under the Electrical Department at Druk Wang Alloys Limited.
- PLC and HMI from Scratch Specialization – Coursera.
- SCADA Software System Basics; Electric Motor Control Systems.
- Artificial Intelligence for Robotics; AI Industry Analysis.
- Business Productivity for Entrepreneurs; Business Entrepreneurship.
- Student Research Meet organised by Royal University of Bhutan.

MANAGEMENT AND OPERATIONS READINESS

- Able to support engineering and facility operations through task coordination, technical reporting, documentation control and follow-up.
- Practical exposure to preventive maintenance observation, safety practices, field checks and operational discipline.
- Comfortable working with engineers, technicians, peers and project teams in site-based and academic project environments.
- Service-focused mindset with awareness of reliability, comfort, safety, response time and smooth technical operations.
- Management-related training supports understanding of leadership, project planning, stakeholder coordination and productivity.

RESEARCH, ACHIEVEMENTS AND LEADERSHIP

- IEEE publication submitted: *Design and Development of a Basic Health Monitoring Station*.
- Participated in Student Research Meet organised by Royal University of Bhutan.

- Winner of college-level Business Idea Competition while representing College of Science and Technology.
- Founder of Zumthruel, a virtual reality gaming venture.
- Class Representative for B.E. Year 1 and Year 2.
- Public Address Councilor for college-wide communication and events.
- Active contributor to Entrepreneur Club, GovTech Satellite Program, Vocational Club and Integrity Club.
- Volunteered in college events, technical activities and student-support roles.

SOFT SKILLS

- Field troubleshooting, practical problem solving and fast learning in industrial plant environments.
- Technical documentation, reporting discipline, drawing interpretation and handover note preparation.
- Team coordination, professional communication and cooperative work with engineers and technicians.
- Safety awareness during electrical, automation, panel assembly, facility support and commissioning work.
- Calm and responsible approach during testing activities, with willingness to take responsibility.
- Service-oriented mindset with focus on reliability, safety, comfort and operational continuity.

ROLE FIT AND CAREER FOCUS

Target Roles	Graduate Engineer, PLC/SCADA Trainee, Instrumentation Trainee, Electrical Automation Engineer, Maintenance Engineer, Facility Operations Support and Commissioning Engineer
Industrial Fit	Hydropower plants, substations, ferro-alloy plants, utilities, manufacturing plants, process industries, facility operations and industrial automation teams
Immediate Value	Commissioning support, PLC/SCADA understanding, loop checking, field wiring verification, signal tracing, panel assembly exposure, documentation and fast site learning
Career Interests	Engineering operations management, facility operations coordination, maintenance planning, team supervision support, hydropower automation, industrial SCADA, electrical maintenance, HVAC control, protection systems, embedded systems and industrial IoT

LANGUAGES

English	Professional working proficiency
Dzongkha	Professional working proficiency
Hindi	Conversational proficiency
Lhotshamkha	Native proficiency
Tshanglakha	Conversational proficiency

REFERENCES

Deepan Sharma	CEO, Bhutan Automation and Engineering Limited Phone: +975 17641448 Email: ceo@bhutanautomation.com
Ashika Chettri	Control Engineer, Bhutan Automation and Engineering Limited Phone: +975 77936493
Karma Kelzang Eudon	Faculty, College of Science and Technology Phone: +975 17607817
Mani Kumar Powdyel	Faculty Member, College of Science and Technology Phone: +975 17369937

DECLARATION

I hereby declare that all the information provided above is true, accurate and complete to the best of my knowledge and belief.